Android Ransomware-like Activity Early Warning in Controlled Lab

A3-PS001-TC185 Team Jackfruit

Shreyas Bhattacharya
bhattacharyashreyas2007@gmail.com

Shrivatsa Kulkarni
shrivatsa.191207@gmail.com

Problem Description

1. Inability of Traditional Tools to Detect Zero-Day Threats: Signature-based antivirus tools fail because the malicious ransomware binary has not yet been cataloged or seen before.

2. Android Application Sandboxing Restrictions: Strict Android security sandboxing prevents traditional security software from directly inspecting the memory or private storage space of other running apps.

3. Disguised Malicious Footprints: Ransomware on Android doesn't broadcast its intent, but leaves a distinctive physical behavioral pattern—specifically, a rapid sequence of file open, write, and rename operations across local storage.

Background

1. Android sandboxing
2. How Android handles files
3. Understanding and handling file telemetry
4. ADB and android emulation - For purposes of testing and prototyping
5. Honeyfile concept

Literature and RFC Review

KRDroid -> Android ransomware file monitoring solution

Lightweight crypto-ransomware detection in Android based on reactive honey file monitoring - JA Gomez-Hernandez, P Garcia-TeodoroSensors, 2024 - mdpi.com

Research into mobile OS security architectures emphasizes that strict application sandboxing prevents security apps from directly inspecting other applications' memory spaces. The above paper advocates for targeted directory monitoring (Taking inspiration from inotify mechanism). Reactive File-System Monitoring (inotify / FileObserver): Utilizes Android's built-in file observation APIs to capture real-time file access and modification telemetry within scoped directories, enabling lightweight behavioral detection without violating application sandboxing constraints.

Proposed Solution

1 .Environment Setup: Build a lightweight, host-based behavioral detection prototype within a controlled, instrumented Android emulator or rooted lab environment.

2. Telemetry Monitoring: Continuously monitor file-system events (open, write, and rename operations) within an explicitly accessible test directory.

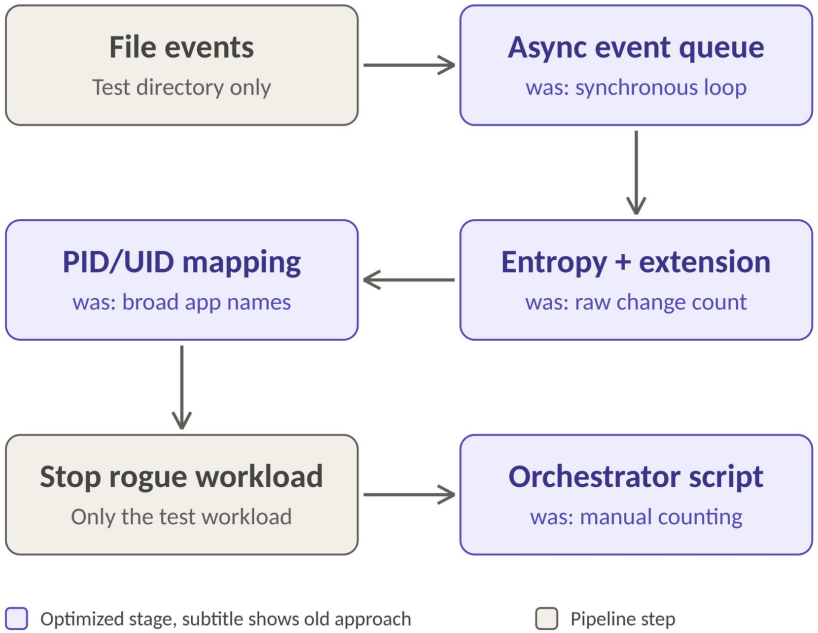
3. Heuristic Analysis: Utilize a sliding-window heuristic engine to identify the rapid I/O anomaly patterns characteristic of unknown or zero-day ransomware.

4. Targeted Mitigation: Instantly generate an early-warning alert and execute a targeted process termination (such as an ADB or root shell command) to stop only the rogue test workload without disrupting the rest of the system.

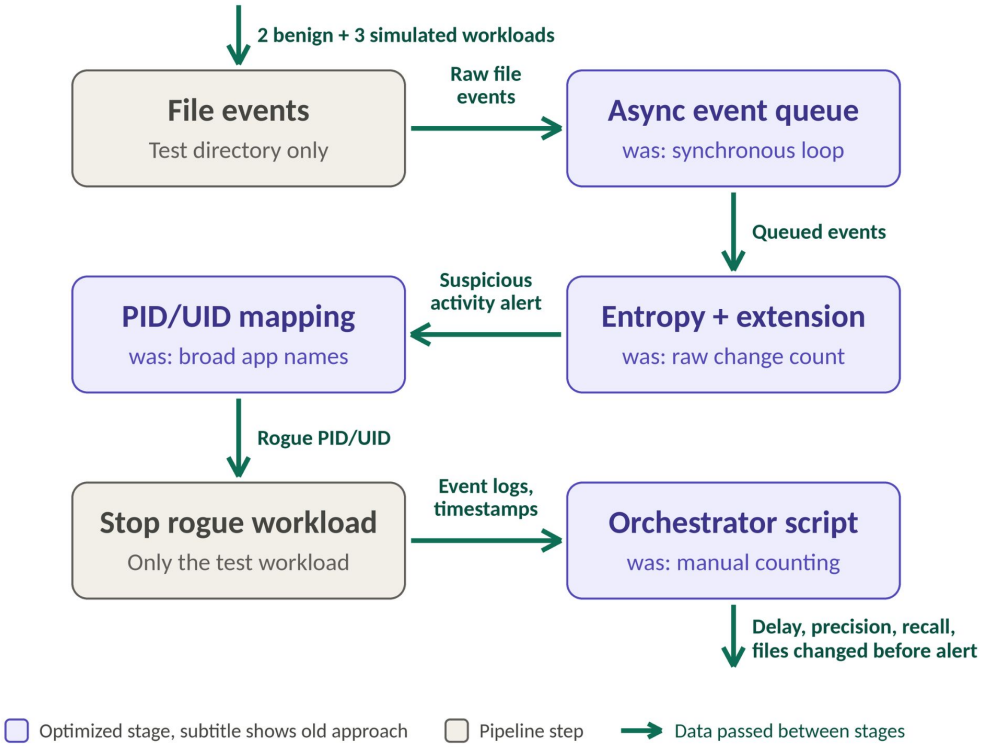
Optimisation Proposed

Enhancement area	Basic approach (before)	Optimized approach (after)
Detection accuracy	Counts raw file changes over time	Adds entropy and extension checking
Performance	Synchronous loop that risks lagging or dropping logs	Asynchronous queue with worker threads
Process control	Broadly targets app names or guesses	Precise PID/UID mapping
Testing and metrics	Manual log counting and script runs	Automated orchestrator script

Solution Architecture & Design



Dataflow


2 benign + 3 simulated workloads
File events
Test directory only
Raw file events
Async event queue
was: synchronous loop
Queued events
Entropy + extension
was: raw change count
Suspicious activity alert
PID/UID mapping
was: broad app names
Rogue PID/UID
Stop rogue workload
Only the test workload
Event logs, timestamps
Orchestrator script
was: manual counting
Delay, precision, recall, files changed before alert
Optimized stage, subtitle shows old approach
Pipeline step
Data passed between stages

Deployment Timeline (3 weeks)

Methodology :

1. Environment Isolation: Testing is strictly isolated to a single instrumented Android emulator or rooted lab device
2. 5-part workload test : Testing 2 benign workloads (bulk unzipping, batch photo saving) and 3 malicious ransomware simulations (rapid mass-rename, in-place rewrite + rename, and multi-threaded burst I/O)

Milestone :

1. Configuring and setting up virtual android environment
2. Implementing confirmed file observation
3. PID UID Mapping
4. Detecting anomaly and killing process
5. Simulation testing and evaluation

Bibliography

Lightweight crypto-ransomware detection in Android based on reactive honeyfile monitoring - JA Gómez-Hernández, P García-Teodoro Sensors, 2024 - mdpi.com

Uncovering the face of android ransomware: Categorisation and Real-time Detection - J Chen, C Wang, Z Zhao, K Chen - ieee.org

Krdroid: ransomware-oriented detector for mobile devices based on behaviors S Wang, S Qin, J Qin, H Zhang, T Tu, Z Jin, J Guo - Applied Sciences, 2021 - mdpi.com

Thank You